THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF MATHEMATICS AND STATISTICS

Final Exam Term 2 2023

MATH3311/Math5335
Computational Mathematics for Finance

(1) TIME ALLOWED – 2 HOURS

(2) READING TIME – 10 MINUTES

(3) THE EXAMINATION PAPER HAS 7 PAGES

(4) TOTAL NUMBER OF QUESTIONS TO BE ANSWERED – 4

(5) ANSWER ALL QUESTIONS

(6) THE QUESTIONS ARE OF EQUAL VALUE

(7) TOTAL NUMBER OF MARKS – 80

(8) START EACH QUESTION ON A SEPARATE BOOK.

(9) THIS PAPER MAY BE RETAINED BY THE CANDIDATE

(10) MATERIAL ALLOWED – ONE A4 PAGE WITH NOTES (YOU CAN WRITE ON BOTH SIDES)

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

Start a new book clearly marked Question 1

1. [20 marks]

Consider the matrix $A = [a_{i,j}] \in \mathbb{R}^{m\times n}$. In this question, the 2-norm is denoted by $\|\cdot\|$.

i) Suppose that $A = QR$ where $Q \in \mathbb{R}^{m\times m}$ is orthogonal and $R \in \mathbb{R}^{m\times n}$ is upper triangular. Show that $\|Q\mathbf{x}\| = \|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^m$.

ii) Assume that $m = n$.

a) State the definition of a positive semidefinite matrix. What extra property is required for a matrix to be (strictly) positive definite?

b) Show that $a_{i,i} > 0$ for all $1 \le i \le n$ when A is (strictly) positive definite.

c) Determine the sparsity of A when $a_{i,i} = 1$ for $1 \le i \le n$, $a_{i,i+1} = 2$ for $1 \le i \le n-1$, while all other remaining entries are zeros.

iii) Assume that $m = n$ and the matrix A is non-singular and has the singular value decompostion, $A = U\Sigma V^T$ where the singular values are ordered as:

$$\sigma_1 \ge \sigma_2 \ge \cdots \ge \sigma_n > 0.$$

a) Show that $\|\Sigma\mathbf{x}\| \le \sigma_1\|\mathbf{x}\|$ for any $\mathbf{x} \in \mathbb{R}^n$.

b) Find a vector $\mathbf{x} \in \mathbb{R}^n$ such that $\|\Sigma\mathbf{x}\| = \sigma_1\|\mathbf{x}\|$.

c) Show that $\|A\| = \sigma_1$, where here the matrix norm is the subordinate matrix norm $\|A\| = \max_{\mathbf{x}\in\mathbb{R}^n\setminus\{\mathbf{0}\}} \frac{\|A\mathbf{x}\|}{\|\mathbf{x}\|}$. You may need to use $A = U\Sigma V^T$, the above parts a) and b), the fact that U and V are orthogonal matrices in $\mathbb{R}^{n\times n}$, as well as part(i).

d) Show that $\|A^{-1}\| = 1/\sigma_n$. (Hint: Note that $A^{-1} = V\Sigma^{-1}U^T$.)

e) State the definition of the condition number of A, cond(A), and then find cond(A) with respect to the 2-norm in terms of the singular values of A.

Start a new book clearly marked Question 2

2. [**20 marks**]

i) Write down a 4×4 permutation matrix P such that

$$P\mathbf{x} = [x_1, x_4, x_2, x_3]^T, \quad \text{for any} \quad \mathbf{x} = [x_1, x_2, x_3, x_4]^T \in \mathbb{R}^4 .$$

ii) Let f be a smooth function and $h > 0$. Consider the expression

$$\frac{3f(x) - 4f(x-h) + f(x-2h)}{2h} = f'(x) - c\,h^2\,f'''(x) + O(h^3).$$

a) Use Taylor series expansions to find the constant c.

b) Derive from the above part a difference approximation of order $O(h^2)$.

iii) Determine the values of the coefficients a, b, c, and d so that

$$s(x) = \begin{cases} s_1(x) = 1 + x - ax^2 + bx^3, & \text{for} \quad x \in [0,1], \\ s_2(x) = c + d(x-1) + (x-2)^3, & \text{for} \quad x \in [1,2]. \end{cases}$$

is a natural cubic spline on $[0, 2]$. Recall that a natural cubic spline satisfies $s''(0) = s''(2) = 0$. (Hint: Start with $s''(0) = s''(2) = 0$ to find the value of the coefficient a first.)

iv) Consider a quadrature formula of the form

$$\int_0^1 g(x)dx \approx w_1\,[g(t) + g(1-t)] + w_2\,g(1/2),$$

where $0 < t < 1/2$, and denote the quadrature error for a given function g by

$$E(g) := \int_0^1 g(x)\,dx - w_1\,(g(t) + g(1-t)) - w_2\,g(1/2)$$

For $k = 0, 1, 2, \ldots$, set

$$E_k = E((x - 1/2)^k).$$

a) Show that $E_0 = 1 - (2w_1 + w_2)$.

b) Show that $E_k = 0$ for $k = 1, 3, 5, \ldots$.

c) Show that

$$E_k = \frac{1}{(k+1)2^k} - 2w_1(t - 1/2)^k \text{ for } k = 2, 4, 6, \ldots .$$

d) Find the values of w_1, w_2 and t so that $E_0 = E_2 = E_4 = 0$.

e) Explain briefly (without proof) why the quadrature rule with t, w_1, w_2 from d) is exact for all polynomials of degree ≤ 5.

Start a new book clearly marked Question 3

3. [**20 marks**]

i) Consider a real-valued random variable Y with a continuous, strictly increasing, cumulative distribution function F, so that

$$P(Y \leq y) = F(y).$$

Let X be a uniformly distributed variable in the interval $[0, 1]$.

a) Show that the random variable $F^{-1}(X)$ has distribution with law F.

b) Suppose we have a procedure to generate samples $x_1, x_2, \ldots$ for X. How can we use these samples to generate samples from Y?
What is likely to be the main practical difficulty with this approach?

ii) The stock price S_t is assumed to follow a geometric Brownian motion

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t, \quad t \in (0, T],$$

where W_t is a standard Brownian motion, μ, σ are positive constants and S_0 is given. Let $N > 1$ be an integer.

a) Describe how can you simulate N asset price paths $(S_{t_0}^{(j)}, S_{t_1}^{(j)}, \ldots, S_{t_R}^{(j)})$, $j = 1, 2, \ldots, N$, at times $0 = t_0 < t_1 < t_2 < \cdots < t_R = T$, where $S_{t_0} = s_0$ is a given value. (The points are not necessarily equally spaced, i.e. it can happen that $t_n - t_{n-1} \neq t_m - t_{m-1}$ for some $m \neq n$.) Hint: You can use the descritization

$$\frac{S_{t_{r+1}} - S_{t_r}}{S_{t_r}} = \mu(t_{r+1} - t_r) + \sigma\sqrt{t_{r+1} - t_r}Z,$$

where Z is a standard normal random variable.

b) Suppose there is a financial derivative with payoff function

$$P = \max\left\{\frac{1}{L}\sum_{\ell=1}^{L}(S_{\tau_\ell} - K), 0\right\},$$

where $0 < \tau_1 < \tau_2 < \ldots < \tau_{L-1} < \tau_L = T$ are given time points (specified by the financial derivative).

A) Outline an algorithm which approximates P.

B) Which parameter(s) do you have to increase to get a more accurate approximation of P?

iii) The Pell numbers P_n, $n \geq 0$, are defined as $P_0 = 0$, $P_1 = 1$, and

$$P_n = 2P_{n-1} + P_{n-2}, \quad n \geq 2.$$

The matrix equation

$$\begin{pmatrix} P_{n+1} & P_n \\ P_n & P_{n-1} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^n = \underbrace{\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}}_{n \text{ times}},$$

provides an alternative way of computing P_n. (You do NOT have to prove this formula.)

a) Outline an algorithm which computes the Pell number P_{2^m+1} in $\mathcal{O}(m)$ operations.

Start a new book clearly marked Question 4

4. [**20 marks**] The value $V(S,t)$ of an option on an underlying asset with price S at time t satisfies the Black–Scholes PDE

$$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV \tag{1}$$

for $0 < S < \infty$ and $0 < t < T$. We assume that the prevailing risk-free interest rate r and the volatility σ are known positive constants. We also assume that the interest compounds continuously.

Recall that a European style **put** option gives the holder the right (but not the obligation) to sell the underlying asset for the strike price K at expiry T.

In this question we will use a so-called 'put option spread', which is a combination of one short position (i.e. short-sell) of a European style put option with strike price K_s and one long position (i.e. purchase) of a European style put option with strike price K_l.

Both options have the same expiry, and the same underlying asset and we assume that $K_l > K_s$.

Let $V_{s,t}$ be the price of the put option in short position and $V_{l,t}$ be the price of the put option in long position at time t.

i) Draw the graph of the pay-off function of the 'put option spread', i.e. the x-axis is the price of the underlying at expiry time and the y-axis is the pay-off of the put option spread at expiry time.

ii) In an arbitrage free market, do we have $V_{l,0} > V_{s,0}$, $V_{s,0} > V_{l,0}$, $V_{l_0} = V_{s,0}$, or none of these?

iii) As the holder of this option spread, do you benefit from an increase in the price of the underlying asset or a decrease in the price of the underlying asset?

iv) The price V of the put option spread at time t is $V = V_{l,t} - V_{s,t}$. Write down the Black-Scholes PDE for the combined put option spread, i.e., the equation which is satisfied by V. (Use r for the risk-free interest rate, σ for the volatility, t for time and S for the price of the underlying at time t.)

v) Carefully explain what the "initial" conditions (i.e. at expiry time T) are for the put option spread.

vi) Carefully explain what the boundary conditions are for the put option spread.

vii) Let $S_{\max} > K_l > K_s$ be the maximal price of the underlying used in the finite difference scheme and $t \in [0,T]$. Set up a grid $(S_n, t_j) = (n\Delta S, j\Delta t)$ for $n = 0, 1, 2, \ldots, N$ and $j = 0, 1, 2, \ldots, J$, where

$$\Delta S = \frac{S_{\max}}{N} \qquad \text{and} \qquad \Delta t = \frac{T}{J}.$$

Let $V_n^j \approx V(S_n, t_j)$ denote the approximate solution. At the time t_j and underlying price S_n give:

a) the **central difference approximation** of $O(\Delta S^2)$ for the partial derivative at stock price S_n and time t_j

$$\frac{\partial V}{\partial S}.$$

b) the **central difference approximation** of $O(\Delta S^2)$ for the partial derivative at stock price S_n and time t_j

$$\frac{\partial^2 V}{\partial S^2}.$$

c) the **backward difference approximation** of $O(\Delta t)$ for the partial derivative at stock price S_n and time t_j

$$\frac{\partial V}{\partial t}.$$

viii) We want to use a finite difference method for computing an approximate solution V_n^j to the Black-Scholes equation (1).

Show that the finite difference equation has the form

$$V_n^{j-1} = \alpha_n V_{n-1}^j + \beta_n V_n^j + \gamma_n V_{n+1}^j,$$

and find the coefficients α_n, β_n and γ_n.

ix) Is this scheme explicit or implicit or neither?

End of Examination

END OF EXAMINATION

Question 1

i) $$\|Q\underline{x}\|^2 = \langle Q\underline{x}, Q\underline{x}\rangle = \underline{x}^T \underbrace{Q^T Q}_{=I} \underline{x} = \underline{x}^T I \underline{x} = \underline{x}^T \underline{x}$$

$$= \|\underline{x}\|^2$$

ii) a) $\underline{x}^T A \underline{x} \geq 0 \quad \forall\, \underline{x} \in \mathbb{R}^n$ ~~pos. def. / or~~ pos. semidef.

$x^T A \underline{x} > 0 \quad \forall\, \underline{x} \in \mathbb{R}^n \setminus \{\underline{0}\}$ strictly pos. def.

b) Let $\underline{e}_i = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow$ position i. Then

$$\underbrace{0 <}_{\text{strictly pos. def.}} \underline{e}_i^T A \underline{e}_i = (0 \ldots 0\, 1\, 0 \ldots 0) \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{i1} & \cdots & a_{in} \\ \vdots & & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

$$= (a_{i1} \cdots a_{in}) \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} = a_{ii}.$$

iii) a)

$$\|\Sigma \underline{x}\|^2 = (\sigma_1 x_1)^2 + (\sigma_2 x_2)^2 + \dots + (\sigma_n x_n)^2$$

$$\leq \sigma_1^2 \left[x_1^2 + x_2^2 + \dots + x_n^2 \right] = \sigma_1^2 \|\underline{x}\|^2$$

b) Let $\underline{x} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$. Then $\Sigma \underline{x} = \begin{pmatrix} \sigma_1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ and $\|\Sigma \underline{x}\| = \sigma_1 = \sigma_1 \underbrace{\|\underline{x}\|}_{=1}$

c) $$\|A\| = \max_{\underline{x} \neq \underline{0}} \frac{\|A\underline{x}\|}{\|\underline{x}\|} = \max_{\underline{x} \neq \underline{0}} \frac{(\underline{x}^T V \Sigma^T U^T U \Sigma V^T \underline{x})^{1/2}}{\|x\|}$$

$$= \max_{\underline{x} \neq 0} \frac{(\underline{x}^T V \Sigma^2 V^T \underline{x})^{1/2}}{\|\underline{x}\|}.$$

Set $\underline{y} = V^T \underline{x}$. Then $\underline{x} = V \underline{y}$ and

$$\|\underline{x}\|^2 = \underline{x}^T \underline{x} = \underline{y}^T \underbrace{V^T V}_{=I} \underline{y} = \underline{y}^T \underline{y} = \|\underline{y}\|^2.$$

Hence

$$\|A\| = \max_{\underline{x} \neq \underline{0}} \frac{(\underline{x}^T V \Sigma^2 V^T \underline{x})^{1/2}}{\|\underline{x}\|} = \max_{\underline{y} \neq 0} \frac{(\underline{y}^T \Sigma^2 \underline{y})^{1/2}}{\|\underline{y}\|}$$

$$= \max_{\underline{y} \neq 0} \frac{\|\Sigma \underline{y}\|}{\|\underline{y}\|} \leq \sigma_1.$$

In fact we get equality when $\underline{y} = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$ and hence $\|A\| = \sigma_1$.

d)

$$\|A^{-1}\| = \max_{\underline{x} \neq \underline{0}} \frac{\|A^{-1}\underline{x}\|}{\|\underline{x}\|}$$

Set $\underline{y} = A^{-1}\underline{x}$. Hence $\underline{x} = A\underline{y}$ and

$$\max_{\underline{x} \neq 0} \frac{\|A^{-1}\underline{x}\|}{\|\underline{x}\|} = \max_{\underline{y} \neq 0} \frac{\|\underline{y}\|}{\|A\underline{y}\|} = \frac{1}{\min_{\underline{y} \neq 0} \frac{\|A\underline{y}\|}{\|\underline{y}\|}}$$

Further

$$\|A\underline{y}\|^2 = \underline{y}^T V \Sigma^T U^T U \Sigma V^T \underline{y} = \underline{y}^T V \Sigma^2 V^T \underline{y}.$$

Set $\underline{z} = \cancel{A}V^T\underline{y}$. Then $\|\underline{z}\| = \|V^T\underline{y}\| = \|\underline{y}\|$ and

$$\|A\underline{y}\|^2 = \underline{z}^T \Sigma^2 \underline{z}. = \|\Sigma \underline{z}\|^2.$$

Thus

$$\min_{\underline{y} \neq 0} \frac{\|A\underline{y}\|}{\|\underline{y}\|} = \min_{\underline{z} \neq 0} \frac{\|\Sigma \underline{z}\|}{\|\underline{z}\|} = \sigma_n,$$

where the last equality follows from

$$\|\Sigma \underline{z}\| \geq \sigma_n$$

and equality follows for $\underline{z} = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$.

e) $\operatorname{cond}(A) = \|A\| \|A^{-1}\| = \frac{\sigma_1}{\sigma_n}$

Question 2

i)

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_4 \\ x_6 \\ x_5 \\ x_2 \\ x_3 \end{bmatrix}$$

ii)

a) $f(x) = f(x)$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2!} f''(x) - \frac{h^3}{3!} f'''(x) + O(h^4)$$

$$f(x-2h) = f(x) - 2hf'(x) + \frac{(2h)^2}{2!} f''(x) - \frac{(2h)^3}{3!} f'''(x) + O(h^4)$$

$$3f(x) - 4f(x-h) + f(x-2h) =$$

$$\cancel{3f(x)} - \cancel{4f(x)} + 4hf'(x) - 4\frac{h^2}{2!} f''(x) + 4\frac{h^3}{3!} f'''(x)$$
$$+ \cancel{f(x)} - 2hf'(x) + \frac{(2h)^2}{2!} f''(x) - \frac{(2h)^3}{3!} f'''(x) + O(h^4)$$
$$= 2hf'(x) - 4\frac{h^3}{3!} f'''(x) + O(h^4)$$

$$\Rightarrow \frac{3f(x) - 4f(x-h) + f(x-2h)}{2h} = f'(x) - \frac{2}{3!} h^2 f'''(x) + O(h^3)$$

$$\Rightarrow C = \frac{2}{3!} = \frac{1}{3}.$$

b) $$f'(x) \approx \frac{3f(x) - 4f(x-h) + f(x-2h)}{2h}$$

iii) $s''(0) = -2a = 0 \Rightarrow a = 0. \Rightarrow s_1(x) = 1 + x + bx^3$

$s''(2) = 6(x-2)\Big|_{x=2} = 0.$

$s_1(1) = 2 + b$ $\qquad$ $s_2(1) = c - 1$

$s_1'(1) = 1 + 3b$ $\qquad$ $s_2'(1) = d + 3$

$s_1''(1) = 6b$ $\qquad$ $s_2''(1) = -6$

$\Rightarrow s_1''(1) = s_2''(1) \Rightarrow 6b = -6 \Rightarrow b = -1$

$\Rightarrow s_1(1) = 2 + b = 1 = s_2(1) = c - 1 \Rightarrow c = 2$

$\Rightarrow s_1'(1) = 1 + 3b = -2 = s_2'(1) = d + 3 \Rightarrow d = -5$

iv) a) $E = \int_0^1 1\,dx - 2w_1 - w_2 = 1 - 2w_1 - w_2$

b) k odd: $\int_0^1 (x - \frac{1}{2})^k dx = 0$

$g(t) + g(1-t) = (t - \frac{1}{2})^k + (1 - t - \frac{1}{2})^k = (t - \frac{1}{2})^k + (-1)^k (t - \frac{1}{2})^k$
$= 0$

$g(\frac{1}{2}) = (\frac{1}{2} - \frac{1}{2})^k = 0.$

$\Rightarrow E = 0.$

k even:

$$\int_0^1 \left(x-\frac{1}{2}\right)^k dx = \frac{1}{k+1}\left(x-\frac{1}{2}\right)^{k+1}\Bigg|_0^1 = \frac{\left(\frac{1}{2}\right)^{k+1} - \left(-\frac{1}{2}\right)^{k+1}}{k+1}$$

$$= \frac{1-(-1)^{k+1}}{(k+1)2^{k+1}} = \frac{1}{(k+1)2^k}$$

$$g(t)+g(1-t) = \left(t-\frac{1}{2}\right)^k + \left(1-t-\frac{1}{2}\right)^k = \left(t-\frac{1}{2}\right)^k\Big(1+\underbrace{(-1)^k}_{=1}\Big)$$

$$= 2\left(t-\frac{1}{2}\right)^k.$$

$$g\left(\frac{1}{2}\right) = \left(\frac{1}{2}-\frac{1}{2}\right)^k = 0.$$

$$\Rightarrow E = \frac{1}{(k+1)2^k} - 2w_1\left(t-\frac{1}{2}\right)^k.$$

c) Need: $k=2\ \&\ k=4$: $\dfrac{1}{(k+1)2^k} - 2w_1\left(t-\frac{1}{2}\right)^k = 0$

$$\frac{1}{3\cdot 4} - 2w_1\left(t-\frac{1}{2}\right)^2 = 0 \Rightarrow w_1\left(t-\frac{1}{2}\right)^2 = \frac{1}{24}$$

$$\frac{1}{5\cdot 16} - 2w_1\left(t-\frac{1}{2}\right)^4 = 0 \Rightarrow w_1\left(t-\frac{1}{2}\right)^4 = \frac{1}{160}$$

$$\Rightarrow \frac{w_1\left(t-\frac{1}{2}\right)^4}{w_1\left(t-\frac{1}{2}\right)^2} = \frac{24}{160} = \left(t-\frac{1}{2}\right)^2$$

$$\Rightarrow \frac{3}{20} = \left(t-\frac{1}{2}\right)^2 \Rightarrow t = \frac{1}{2} \pm \sqrt{\frac{3}{20}} \Rightarrow t = \frac{1}{2} - \sqrt{\frac{3}{20}}$$

since $0 < t < \frac{1}{2}$.

$$\Rightarrow w_1 = \frac{1}{\left(t-\frac{1}{2}\right)^2 24} = \frac{20}{3}\,\frac{1}{24} = \frac{5}{2\cdot 9}.$$

For $g(x) = 1$ we get

$$0 = E = 1 - 2w_1 - w_2 \Rightarrow w_2 = 1 - 2w_1 = \frac{4}{9}.$$

e) Any polynomial p of degree ≤ 5 can be written in the form

$$p(x) = a_0 + a_1\left(x-\frac{1}{2}\right) + \dots + a_5\left(x-\frac{1}{2}\right)^5.$$

Then

$$E(p) = a_0 E_0 + a_1 E_1 + \dots + a_5 E_5 = 0.$$

Question 3

i) a) $\mathbb{P}(F^{-1}(X) \le y) = \mathbb{P}(X \le F(y)) = F(y)$ since X is uniformly distributed.

b) The samples $F^{-1}(x_1), F^{-1}(x_2), \dots$ are samples from Y.

The main difficulty is likely the computation of F^{-1}.

ii) a) Using the discretization

$$\frac{\Delta S_n}{S_n} = \frac{S_{n+1} - S_n}{S_n} = \mu \Delta t_n + \sigma \sqrt{\Delta t_n}\, Z_n,$$

where $Z_n \sim N(0,1)$ and $\Delta t_n = t_{n+1} - t_n$.

Thus

$$S_{n+1} = S_n\left(1 + \mu \Delta t_n + \sigma \sqrt{\Delta t_n}\, Z_n\right).$$

```
t given
S = zeros(N, R+1);

S(:, 1) = s_0;        (this is Matlab convention; Python: start index with (
Z = randn(N,R)
for k = 1:N
    for n = 2:R+1
        S(k,n) = S(k,n-1)·(1+μ·(t(n+1)-t(n))
                 + σ·sqrt(t(n+1)-t(n)) · Z(k,n-1))
    end
end
```

b)

A) Apply part a) with the time points

$$0 = t_0 < t_1 = \tau_1 < \dots < t_L = \tau_L = T \quad \text{with } R = L.$$

Use simulation from part a) to get array S of size $N \times (R+1)$.

```
Papp = 0
for k = 1 : N
    Papp = Papp + max(0, 1/L sum(S(k, 2:L+1) - K));
end
Papp = Papp / N;
```

B) Increase N;

Note that L is fixed and cannot be changed.

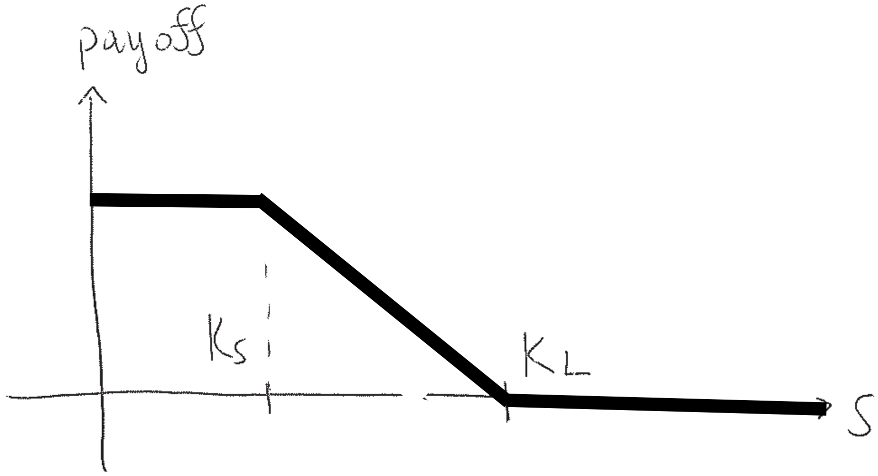
iii)

```
A = [2, 1; 1, 0];
for k = 1 : m
    A = A * A;
end
```

$P_{2^m+1} = A(1,1);$

Question 4

i)



ii) $V_{L,0} > V_{S,0}$

iii) decrease

iv) $$\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{\sigma^2}{2}S^2\frac{\partial^2 V}{\partial S^2} = rV$$

v) 'Initial' conditions at expiry time:

$$V(S_T, T) = \max(0, K_\ell - S_T) - \max(0, K_s - S_T).$$

vi)

$$V(S,t) = 0 \text{ as } S \to \infty.$$

$$V(0,t) = (K_\ell - K_s)e^{-r(T-t)}$$

vii) a)

$$\frac{\partial V}{\partial S}(S_n, t_j) = \frac{V_{n+1}^j - V_{n-1}^j}{2\Delta S}$$

b)

$$\frac{\partial^2 V}{\partial S^2}(S_n, t_j) = \frac{V_{n+1}^j - 2V_n^j + V_{n-1}^j}{\Delta S^2}$$

c)

$$\frac{\partial V}{\partial t}(S_n, t_j) = \frac{V_n^j - V_n^{j-1}}{\Delta t}$$

viii)

$$\frac{V_n^j - V_n^{j-1}}{\Delta t} + r S_n \frac{V_{n+1}^j - V_{n-1}^j}{2\Delta S} + \frac{\sigma^2}{2} S_n^2 \frac{V_{n+1}^j - 2V_n^j + V_{n-1}^j}{\Delta S^2}$$

$$= r V_n^j$$

$$V_n^{j-1} = V_n^j - \Delta t\, r V_n^j + r\Delta t S_n \frac{V_{n+1}^j - V_{n-1}^j}{2\Delta S} + \frac{\sigma^2}{2}\Delta t S_n^2 \frac{V_{n+1}^j - 2V_n^j + V_{n-1}^j}{\Delta S^2}$$

$$V_n^{j-1} = V_{n-1}^j \underbrace{\left(\Delta t \frac{\sigma^2}{2} \frac{S_n^2}{\Delta S^2} - \frac{r\Delta t}{2} \frac{S_n}{\Delta S}\right)}_{\alpha_n}$$

$$+ V_n^j \underbrace{\left(1 - \Delta t r - \Delta t \frac{\sigma^2}{2} \frac{S_n^2}{\Delta S^2} \cdot 2\right)}_{\beta_n}$$

$$+ V_{n+1}^j \underbrace{\left(\frac{r\Delta t}{2} \frac{S_n}{\Delta S} + \frac{\sigma^2}{2}\Delta t \frac{S_n^2}{\Delta S^2}\right)}_{\gamma_n}$$

ix) explicit